RealScope : a dataset of lab manual diagrams and microscope images for 2nd PU Biology

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Abstract

2nd PU Biology students learn microscopy from line diagrams in a printed lab manual, but the slides they see under the microscope look very different. We present RealScope, a dataset that pairs 53 labelled diagrams from the II PUC Biology lab manual with a supplied microscope image of the same specimen, and a static website built from it. Each record links the diagram crop, its printed page, its labels, the image source and the licence text supplied with it. Diagrams were cut from the scanned manual with a reproducible Python pipeline. The dataset, code and website are openly available.

Background & Summary

2nd PU Biology students prepare for the practical exam from the small black-and-white diagrams in their lab manual. A stained slide under the microscope looks very different, so students often mistake things like air bubbles or dust for cell parts. RealScope places each manual diagram next to a real microscope image of the same specimen: 53 specimens from 14 manual pages, covering cell division, plant reproduction, pond-water organisms, animal tissues and parasites.

Methods

The pipeline is written in Python. One command, `python Scripts/run_all.py`, runs every step and opens the website.

1. **Specimen list.** Compiled in `specimens.csv`: for each specimen, its manual page, name, magnification, one image link and the licence given by its source.
2. **Diagram extraction.** Each drawing on the 14 pages is located with OpenCV, its caption read with Tesseract OCR, and it is cropped with all its labels kept whole.
3. **Matching.** Each CSV row is matched to a diagram on the same page by name, not by row order, so reordering or deleting rows cannot pair the wrong images.
4. **Real images.** Each link is downloaded once and saved; later runs reuse the saved copy. Logos, error pages and very small images are rejected. If a download fails, the website loads the image directly from its link.

5. **Website.** A static website is built from the matched records, with a label-free copy of each diagram for self-testing.

Data Records

The repository holds six records under `Data/`.

Record	Location	Content
Specimen list	`input/specimens.csv`	53 rows: page, figure, magnification, image link, licence
Diagram metadata	`raw_data/images_metadata.csv`	One row per crop: page, caption, labels and crop position
Diagram crops	`raw_data/extracted_images/`	53 diagrams and 8 grouped sets, PNG
Matching report	`specimen_matching.csv`	CSV row, matched diagram, method, score
Image manifest	`real_slide_images/photos_manifest.csv`	Supplied link, resolved link, status, size, licence
Website	`website/`	`index.html`, data file, display images, 93 manual pages

Technical Validation

Automated checks on the final run (4 November 2026):

Check	Result
CSV rows matched to a diagram by page and name	53 of 53
Same pairings after shuffling the rows and deleting 7	Yes
Website: every diagram readable, every manual-page link valid	53 of 53

Each specimen has its own image link, and the pipeline holds back any image whose link is shared by two specimens, so no image is shown under the wrong name.

Usage Notes

Open `Data/website/index.html` in a browser. The site has four views: **Specimens** (search and filter by topic), **Slide viewer** (diagram and real image side by side, with zoom and a label on/off switch), **Mistakes** (13 common slide artefacts, shown as simulated images) and **Lab manual** (all 93 pages). To change a specimen, edit the CSV and run the command again; only the changed steps rerun.

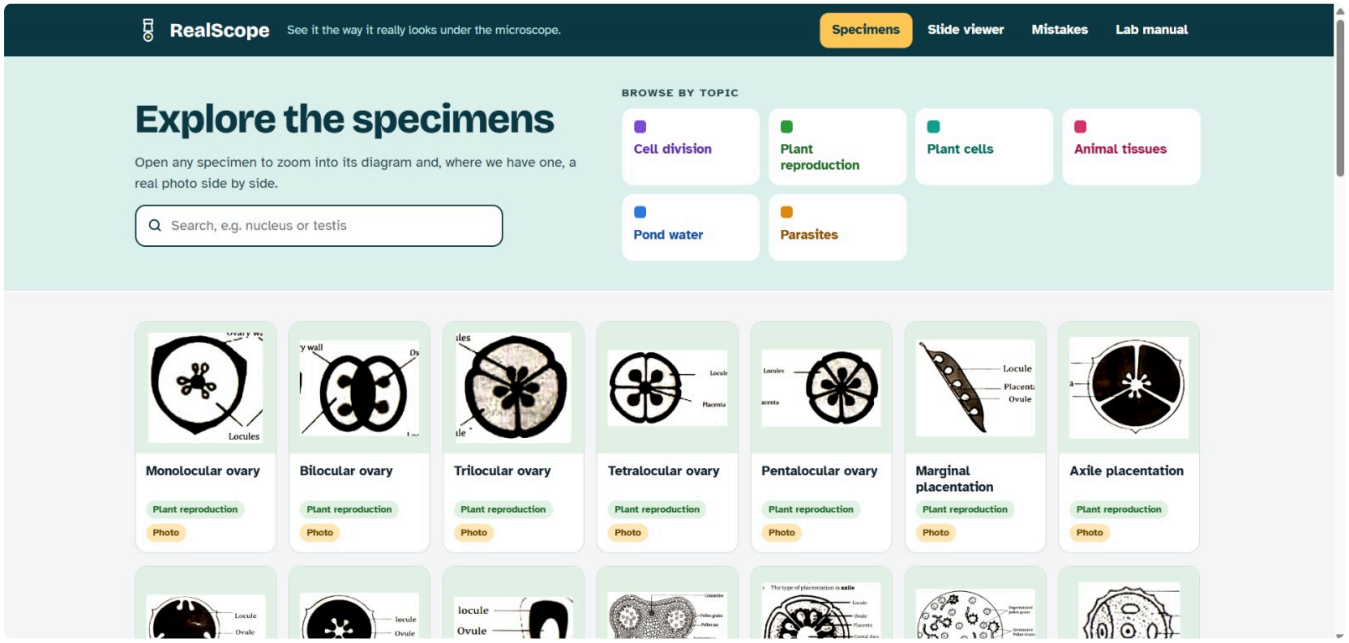


Figure 1. Specimen gallery: search box, topic filters and one card per specimen.

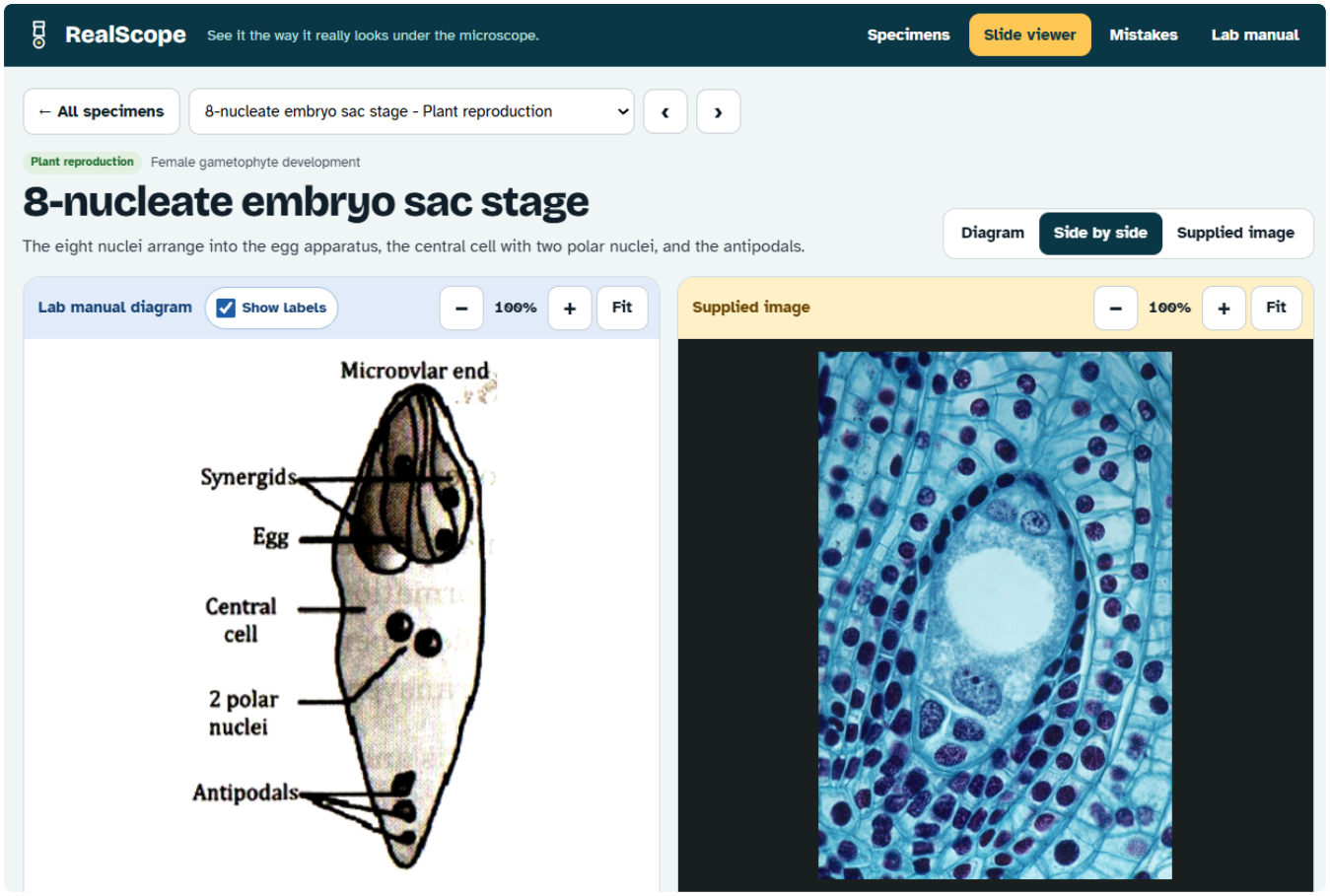


Figure 2. Slide viewer: the lab-manual diagram of the 8-nucleate embryo sac beside a real microscope image of the same specimen, with zoom, a labels switch and diagram / side-by-side / image views.

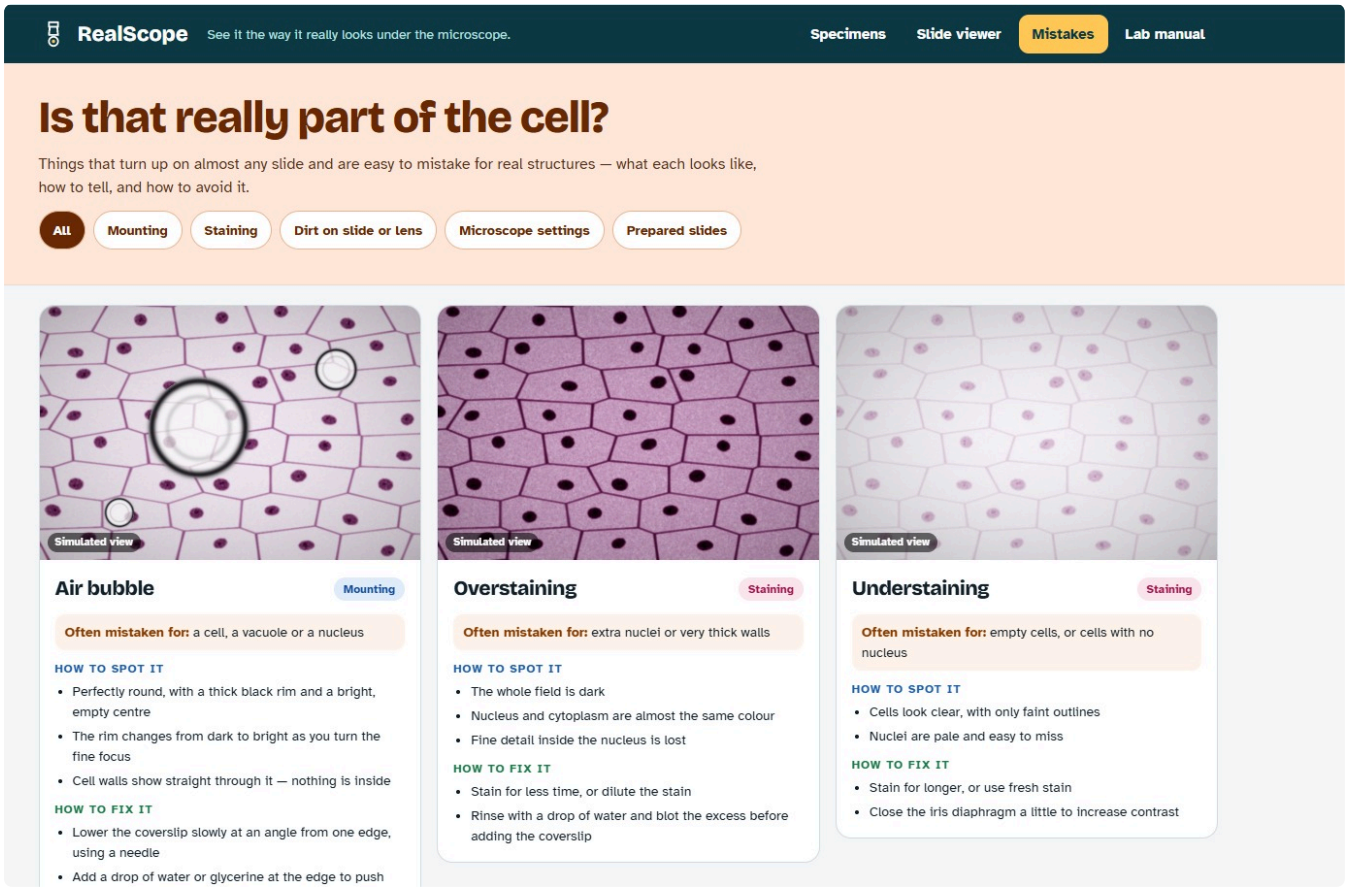


Figure 3. Mistakes guide: simulated views of common slide artefacts, filtered by cause, with how to spot and fix each.

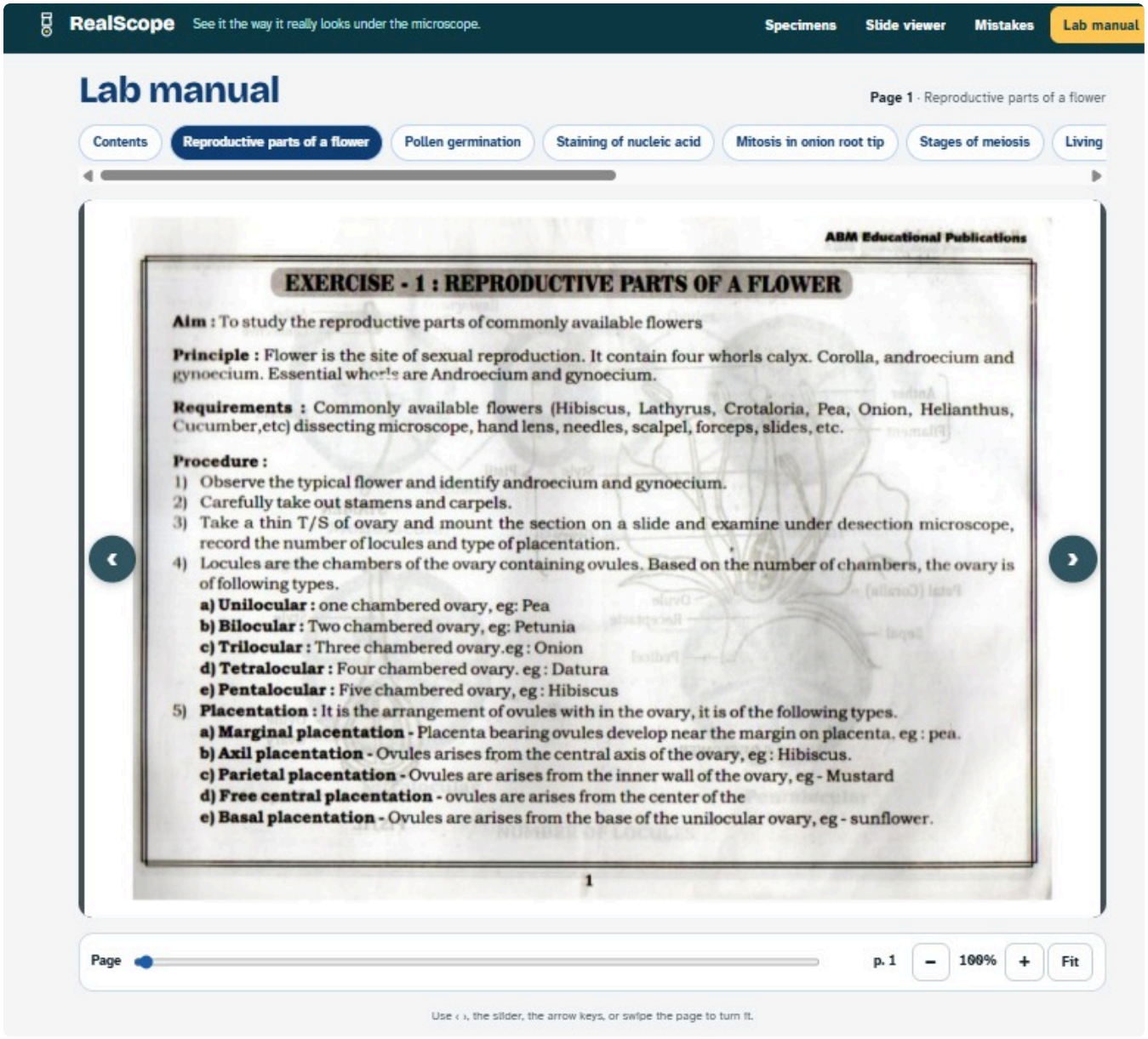


Figure 4. Lab-manual reader: exercise shortcuts, page arrows, page slider and zoom over the scanned manual.

Data Availability

All records are at github.com/BN-Mohith/RealScope. The real images stay under their original licences, listed in `specimens.csv`.

Code Availability

The code is in `Scripts/` in the same repository. It needs Python 3.9 or later, the packages in `requirements.txt` and the Tesseract OCR program.

References

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3. Bradski, G. The OpenCV Library. *Dr. Dobb's Journal of Software Tools* (2000).
4. Smith, R. An overview of the Tesseract OCR engine. *Proc. ICDAR* 629–633 (2007).
5. Scientific Data. Submission guidelines. nature.com/sdata/submission-guidelines

Author Contributions

BN.MOHITH built the pipeline and the website. ATIF K collected the microscope images and their licences and compiled the specimen CSV.